

200 REAL WORLD DEVOPS SCENARIOS

The Unknown Unknowns Map

What You Do Not Know You Do Not Know

Every scenario a DevOps engineer faces in production

Across 10 domains: Linux | Git | Jenkins | Docker | Kubernetes AWS | Monitoring | Security | Terraform |
Networking

For each scenario: What you see | What topic it tests | What you need to know

Use this as a self-assessment. If a scenario surprises you - that is your next learning target.

How to Use This Document

This is not a document to read start to finish. It is a map of everything you might face in DevOps. Go through each scenario and ask yourself honestly:

"Do I know what this scenario means and how to handle it?"

If yes: move on. If no: that is your next learning target. Mark it. Study it. Build it.

THE KEY INSIGHT: The scenarios that surprise you the most are the most valuable. They represent gaps you did not know existed — exactly like the multi-agent question at Citizens Bank.

Level Guide

Level	Who Handles It	What It Means
L1	Junior / L1 DevOps	Follow runbook, monitor, escalate. You should know these NOW.
L2	Mid / L2 DevOps	Diagnose and fix. Build pipelines. You should learn these within 6 months.
L3	Senior / Architect	Design systems, prevent problems. Learn these over 1-2 years.

The 5 Patterns That Repeat Across ALL Tools

RESOURCE EXHAUSTION

Disk full, memory full, CPU maxed. Same problem in Jenkins, Docker, K8s, EC2. Same fix: monitor, alert at 80%, clean or scale.

VERSION MISMATCH

Java 17 vs 21, Node 16 vs 18, Terraform state vs binary. Same fix: pin versions, upgrade together, test before production.

AUTHENTICATION FAILURE

Expired tokens, wrong credentials, permission denied. Same fix: credential management, rotation, IAM roles over hardcoded keys.

SCALING UNDER LOAD

400 Jenkins jobs, 1000 HTTP requests, 50 pods. Same fix: horizontal scaling, auto-scaling, load balancing.

RECOVERY FROM FAILURE

Jenkins crash, container death, pod restart, instance failure. Same fix: self-healing automation (systemd, K8s, ASG).

DOMAIN 1

Linux & Operating System

The foundation everything runs on

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	Server is responding slowly, users complain	CPU/Memory diagnosis	L1	Check top/htop, identify which process consumes most CPU or RAM. Kill or restart it.
2	Disk is 95% full, services starting to fail	Disk management	L1	df -h to find full partition, du -sh to find largest dirs, clean logs/tmp/old builds.
3	Cannot SSH into EC2 — "Connection refused"	SSH troubleshooting	L1	Check: is instance running? Security group port 22 open? Key permissions chmod 400?
4	Process is running but port is not accessible	Port binding	L1	netstat -tlnp: is process listening? On 127.0.0.1 (local only) or 0.0.0.0 (all)?
5	Service starts then immediately crashes	Service debugging	L1	journalctl -u service-name: read the error. Common: wrong config, missing file, port conflict.
6	Need to find which process is using port 8080	Process management	L1	lsof -i :8080 or netstat -tlnp grep 8080. Shows PID. Kill with kill -9 PID.
7	Cron job is not running at scheduled time	Cron scheduling	L2	crontab -l to list. Check: timezone, path issues, permissions, log output to file.
8	Server rebooted unexpectedly overnight	System logs	L2	dmesg, /var/log/syslog, last reboot. Was it OOM killer? Kernel panic? AWS maintenance?
9	File permissions block a service from reading config	Linux permissions	L1	ls -la to check owner. chown/chmod to fix. Understand: user, group, other, rwx.
10	Environment variables set in terminal are gone after reboot	Shell profiles	L1	Variables in terminal are session-only. Put them in ~/.bashrc or /etc/environment for persistence.
11	Running out of file descriptors under high load	System limits	L2	ulimit -n shows limit. Edit /etc/security/limits.conf. Real production issue for busy servers.
12	Need to transfer a file between two EC2 instances	File transfer	L1	scp file.txt user@ip:/path/ or rsync for large/repeated transfers.
13	DNS resolution fails: "Could not resolve host"	DNS	L2	Check /etc/resolv.conf, try nslookup/dig. Check VPC DNS settings in AWS.
14	Server clock is wrong, SSL certificates fail	Time sync	L2	timedatectl to check. Install chrony or ntpd. Clock drift breaks TLS, Kerberos, logs.
15	Need to run a command on 50 servers simultaneously	Automation	L3	Ansible ad-hoc command: ansible all -m shell -a "command". Or use AWS SSM Run Command.

DOMAIN 2

Git & Source Control

How code flows in teams

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	Two developers edited the same file, merge conflict	Merge conflicts	L1	Open file, find <<<< ===== >>>> markers, choose correct code, git add, git commit.
2	Developer accidentally pushed secrets to GitHub	Secret exposure	L1	Rotate the secret IMMEDIATELY. Remove from history with git filter-branch or BFG. Add to .gitignore.
3	Need to undo last commit but keep changes	Git reset	L1	git reset --soft HEAD~1 (keeps staged). --mixed (unstaged). --hard (deletes everything).
4	Feature branch is 200 commits behind main	Branch drift	L2	git rebase main (rewrite history, cleaner) or git merge main (merge commit, safer for shared branches).
5	Release needs one specific fix from another branch	Cherry-pick	L2	git cherry-pick . Applies one commit without merging entire branch.
6	Build works locally but fails on Jenkins — file missing	Gitignore issue	L1	File is in .gitignore so never pushed to GitHub. Jenkins clones from GitHub, file missing.
7	Need to find which commit introduced a bug	Git bisect	L2	git bisect start, mark good/bad commits, Git binary-searches to find the exact breaking commit.
8	Developer force-pushed and overwrote team changes	Force push damage	L2	Recover with git reflog on affected machines. Enforce branch protection rules to prevent.
9	Release branch needs different config than develop	Branch strategy	L2	Gitflow: release/* branch from develop, only bugfixes allowed, merges to main AND develop.
10	100+ developers all committing to same repo	Monorepo management	L3	Trunk-based development, feature flags, path-based CI triggers, CODEOWNERS file.
11	Need to tag a release and deploy that exact version	Git tags	L1	git tag -a v1.0.5 -m "Release 1.0.5" && git push origin v1.0.5. Jenkins triggers on tag.
12	Hotfix needed for production but develop has new features	Hotfix workflow	L2	Branch from main: hotfix/fix-name, fix, merge to main AND develop. Tag main.
13	Pull request has been open for 2 weeks, massive diff	PR best practices	L2	Small PRs (under 400 lines). Break large features into smaller PRs. Review daily.
14	Someone committed a 500MB file to the repo	Large file handling	L2	git filter-branch to remove. Use Git LFS for large files. Add size limits to pre-push hooks.

DOMAIN 3

Jenkins & CI/CD

The orchestration layer

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	400 jobs queued, only 10 agents running	Agent scaling	L2	Add agents, increase executors, dynamic agents (K8s/EC2 plugin), priority sorter, parallel stages.
2	Agent went offline overnight	Agent health	L1	Check SSH/JNLP connection, disk space, Java process, EC2 status. Bring back online.
3	Build stuck for 3 hours, not progressing	Stuck builds	L1	Check console output last line. Kill build. Check for input step, network timeout, deadlock.
4	All builds fail with "401 Unauthorized" at Nexus push	Credential expiry	L1	Nexus password/token expired. Update in Jenkins Credentials store. Re-run build.
5	Jenkins disk full, UI extremely slow	Disk management	L1	Delete old workspaces, enable build discarder, docker system prune on agents.
6	Plugin update broke Jenkins	Plugin management	L2	Downgrade plugin. Or remove from /var/lib/jenkins/plugins/ and restart. Test on staging first.
7	Webhook not triggering builds after GitHub push	Webhook debug	L1	Check GitHub webhook deliveries. Verify URL. Check Jenkins security settings.
8	Different teams need different pipeline standards	Shared Libraries	L2	Create Jenkins Shared Library with standard stages. Teams call library functions.
9	Need approval before production deployment	Approval gates	L2	input{} step in Jenkinsfile. Submitter restrictions. CAB process integration.
10	Builds take 20 minutes, developers frustrated	Build optimization	L2	Parallel stages, Maven caching, Docker layer caching, skip unnecessary stages on branches.
11	Jenkins controller crashed, need to restore	Backup/recovery	L2	Restore from backup: /var/lib/jenkins/ (jobs, config, secrets, plugins). Or use JCasC.
12	Need same pipeline for 50 microservices	Pipeline templates	L3	Shared Library with template pipeline. Each repo has 3-line Jenkinsfile calling library function.
13	Security team wants to enforce scans on all builds	DevSecOps enforcement	L3	Mandatory stages in Shared Library. Admin-only write access. Developers cannot bypass.
14	Jenkins needs to run different tools on different agents	Label routing	L1	Labels on agents (java, docker, windows). Jenkinsfile agent { label 'java' } routes correctly.
15	Need to migrate Jenkins to new server	Jenkins migration	L3	Backup /var/lib/jenkins/, restore on new server. Or use JCasC for config-as-code approach.
16	Developers requesting admin access to Jenkins	RBAC	L2	Matrix authorization or Role-based strategy. Folder-level permissions per team. No global admin.
17	Build passes on one agent, fails on another	Environment drift	L2	Different tools installed on different agents. Standardize with Docker agents or golden AMI.

18	Need to run a job on schedule, not on push	Scheduled builds	L1	triggers { cron('H 2 * * 1-5') } in Jenkinsfile. Or configure in Jenkins job settings.
19	Pipeline needs to call an external API	API integration	L2	Use httpRequest step or sh 'curl...' with credentials. Handle response, fail on error codes.
20	Multiple branches need different deploy targets	Multi-branch	L2	when { branch 'main' } for prod, when { branch 'develop' } for staging. Multi-branch pipeline job.

DOMAIN 4

Docker & Containers

Application packaging

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	Docker image is 2GB, takes 10 min to push	Image size	L2	Multi-stage build. Use slim/alpine base. Remove build tools from final image. Layer caching.
2	"No space left on device" during docker build	Docker disk	L1	docker system prune -a. Clean dangling images, stopped containers, build cache.
3	Container starts then immediately exits	Container debugging	L1	docker logs . docker run -it bash to get shell. Check entrypoint/cmd.
4	Container cannot connect to database on same host	Docker networking	L2	Containers on same docker network can talk by name. Use docker network create + --network flag.
5	Need to pass database URL to container without hardcoding	Environment config	L1	docker run -e DB_URL=... or --env-file .env. Never bake env-specific config into images.
6	Data is lost when container restarts	Docker volumes	L1	Use -v /host/path:/container/path or named volumes. Data persists across container restarts.
7	docker build takes 15 minutes even for small code change	Layer caching	L2	Order Dockerfile: stable (base, deps) first, changing (code) last. Dependencies before source.
8	"Permission denied" when container writes to mounted volume	Volume permissions	L2	Container runs as root but host dir owned by ubuntu. Match UIDs or use named volumes.
9	Need to run a container that auto-restarts on crash	Restart policies	L1	--restart unless-stopped or --restart always. Survives daemon restart and container crash.
10	Production image has security vulnerabilities	Image scanning	L2	Trivy/Grype scans image for CVEs. Integrate into CI pipeline. Block deploys with critical CVEs.
11	Need to run multiple containers that work together	Docker Compose	L2	docker-compose.yml defines multiple services, networks, volumes. docker-compose up starts all.
12	"OOMKilled" — container keeps getting killed	Resource limits	L2	docker run --memory=512m --cpus=1.5. Without limits, container eats all host resources.
13	Docker Hub rate limit: too many pulls	Registry management	L2	Mirror images to private registry (ECR, Harbor). Cache base images. Use authenticated pulls.
14	Need to copy a file from running container to host	Container operations	L1	docker cp ./path/file ./local/file. Works both directions.
15	Build argument needs to be different for dev vs prod	Build args	L2	docker build --build-arg ENV=prod. In Dockerfile: ARG ENV. Avoid for secrets (visible in history).
16	"COPY failed: file not found" during docker build	Build context	L1	Docker only sees files in the build context (directory you run docker build from). Check .dockerignore.

17	Container cannot reach external API on the internet	Container networking	L2	Check DNS (--dns flag), check proxy settings, check security group outbound rules.
18	Need to update one container without affecting others	Rolling updates	L2	In Docker Compose: docker-compose up -d --no-deps . In K8s: rolling update default.
19	Logs from container are filling up the disk	Log management	L2	Configure log driver: --log-opt max-size=10m --log-opt max-file=3. Rotate logs automatically.
20	Need to debug network between containers	Network debugging	L2	docker network inspect. docker exec ping/curl other. Check DNS resolution between containers.

DOMAIN 5

Kubernetes

Container orchestration at scale

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	Pod is in CrashLoopBackOff state	Pod debugging	L1	kubectl logs (and --previous). kubectl describe pod . Check: image, config, resources.
2	Pod cannot pull image: "ImagePullBackOff"	Image pull issues	L1	Wrong image name/tag, private registry without imagePullSecret, or registry down.
3	Application is slow, need more instances	Horizontal scaling	L1	kubectl scale deployment myapp --replicas=5. Or configure HPA for auto-scaling on CPU.
4	New deployment caused errors, need to rollback	Rollback	L1	kubectl rollout undo deployment/myapp. Instant. Previous ReplicaSet becomes active.
5	Pod is running but service returns "connection refused"	Service debugging	L2	Check: service selector matches pod labels? Port correct? kubectl get endpoints shows IPs?
6	Pod is evicted: "The node was low on resource"	Resource management	L2	Node ran out of disk/memory. Set resource requests and limits on pods. Monitor node capacity.
7	Need different config for dev vs production pods	ConfigMaps/Secrets	L1	ConfigMap for non-sensitive. Secret for sensitive. Mount as env vars or files. Different per namespace.
8	DNS resolution fails between services	Cluster DNS	L2	Check CoreDNS pods running. Service name format: ..svc.cluster.local.
9	Need to expose service to internet	Ingress	L2	Create Ingress resource with rules. Requires Ingress controller (nginx, ALB). TLS termination.
10	Node is NotReady, pods rescheduling	Node troubleshooting	L2	kubectl describe node. Check: kubelet running? Disk pressure? Memory pressure? Network?
11	Need to update application with zero downtime	Rolling updates	L1	Default K8s behavior. Deployment creates new pods, waits for ready, then removes old pods.
12	Container needs to read a mounted config file	Volume mounts	L1	Mount ConfigMap as volume. Pod sees it as a file at the specified path.
13	Need to run a one-time database migration	K8s Jobs	L2	Use Job resource (not Deployment). Runs once to completion. Or use init containers.
14	Pod needs to access AWS services (S3, ECR)	IAM for pods	L2	IRSA: IAM Roles for Service Accounts. Pod assumes an IAM role via its ServiceAccount.
15	Need to limit which pods can talk to each other	Network policies	L3	NetworkPolicy resources. Default: all pods can talk. Policy restricts by label, namespace, port.
16	Helm deployment fails: "timeout waiting for condition"	Helm debugging	L2	Pods not becoming Ready. Check: kubectl get pods, kubectl describe pod, kubectl logs.
17	Need to run stateful application (database) in K8s	StatefulSets	L3	StatefulSet: stable pod names, ordered startup, persistent volumes per pod. Not Deployment.

18	Cluster needs to handle sudden traffic spike	Auto-scaling	L3	HPA scales pods. Cluster Autoscaler scales nodes. Karpenter for faster node provisioning.
19	Need to store database passwords for pods	Secret management	L2	K8s Secrets (base64, not encrypted by default). Better: External Secrets Operator + Vault/AWS SM.
20	Pod logs need to be searchable across all services	Centralized logging	L2	EFK/ELK stack or Loki. Fluent Bit as DaemonSet collects logs. Kibana/Grafana for search.

DOMAIN 6

AWS & Cloud

Infrastructure that everything runs on

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	EC2 public IP changed after stop/start	Elastic IP	L1	Use Elastic IP (static) or internal DNS. Public IPs are ephemeral by default.
2	Monthly AWS bill suddenly doubled	Cost management	L2	Check Cost Explorer. Common: forgotten running instances, over-provisioned, unused EBS volumes.
3	Cannot access EC2: "Permission denied (publickey)"	SSH access	L1	Wrong key file, wrong username (ubuntu vs ec2-user), key permissions not 400.
4	Application on EC2 cannot access S3	IAM permissions	L1	Attach IAM Role to EC2 with S3 access policy. Never use hardcoded access keys.
5	Need to serve 10,000 requests per second	Load balancing	L2	ALB distributes traffic to multiple EC2s. ASG adds instances when load increases.
6	EC2 instance failed health check, service down	Auto recovery	L2	ASG detects unhealthy instance, terminates it, launches replacement automatically.
7	Secrets hardcoded in application config files	Secrets management	L2	AWS Secrets Manager or Parameter Store. Application reads at runtime. Rotate automatically.
8	Database is on same EC2 as application	Architecture	L2	Move to RDS (managed DB). Separate compute from data. Backups, failover handled by AWS.
9	VPC has no internet access for private subnets	VPC networking	L2	NAT Gateway in public subnet. Route table in private subnet points to NAT for internet.
10	S3 bucket accessible to entire internet	S3 security	L1	Check bucket policy and ACLs. Block public access. Enable encryption. Audit with Access Analyzer.
11	Need same infrastructure in dev, staging, production	IaC	L3	Terraform modules. Same code, different variable files per environment.
12	Cross-account access needed for shared services	AWS Organizations	L3	IAM cross-account roles. Organizations for multi-account strategy. SCPs for guardrails.
13	Application needs to handle traffic spikes for events	ASG + ALB	L2	Pre-warm ALB. Configure ASG with predictive scaling. Target tracking on CPU/request count.
14	CloudWatch alarm firing but nobody noticed	Alerting	L1	CloudWatch alarms -> SNS -> email/Slack/PagerDuty. Monitor that alerts are being received.
15	Need to encrypt data at rest and in transit	Encryption	L2	At rest: EBS encryption, S3 SSE, RDS encryption. In transit: TLS/HTTPS, ACM certificates.
16	EC2 instance was terminated accidentally	Protection	L1	Enable termination protection. Use lifecycle policies in ASG. Tag critical instances.

17	Need to give temporary access to a contractor	IAM	L2	Create IAM user with time-limited credentials. Or assume role with session duration. MFA required.
18	Application needs to work across two AWS regions	Multi-region	L3	Route 53 for DNS failover. S3 cross-region replication. RDS read replicas. Complex and expensive.
19	EBS volume too slow for application	Storage performance	L2	gp3 vs io2 vs io1. IOPS and throughput matter. Monitor with CloudWatch EBS metrics.
20	CloudFormation stack stuck in UPDATE_ROLLBACK_FAILED	IaC debugging	L3	Manual resource cleanup needed. Check CloudFormation events for the specific resource that failed.

DOMAIN 7

Monitoring & Observability

Seeing what is happening in production

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	Service is "up" but users report errors	Monitoring gap	L1	Monitor user-facing symptoms (error rate, latency) not just server health (CPU, disk).
2	50 alerts fire daily, team ignores all of them	Alert fatigue	L2	Reduce to symptom-based alerts only. Remove cause-based alerts (CPU high). Make every alert actionable.
3	Cannot find what caused an error 3 days ago	Log retention	L2	Centralize logs (ELK/Splunk/Datadog). Set retention policy (30-90 days). Structured logging.
4	Request is slow but don't know which service is the bottleneck	Distributed tracing	L2	Jaeger/Datadog APM traces requests across services. Shows exactly where time is spent.
5	Need to know if we are meeting our reliability target	SLOs/SLIs	L3	Define SLO: "99.9% of requests succeed within 200ms." Measure SLI. Track error budget.
6	Application error but no useful information in logs	Structured logging	L2	Log in JSON format with context: request_id, user_id, action, error. Not just "Error occurred."
7	Dashboard shows everything green but users are unhappy	Wrong metrics	L2	Monitor the 4 Golden Signals: latency, traffic, errors, saturation. Not just CPU/memory.
8	Need to know which deployment caused error spike	Deployment markers	L2	Add deployment events to monitoring dashboards. Correlate deploys with metric changes.
9	On-call engineer missed an alert at 3am	On-call process	L1	PagerDuty/Opsgenie with phone call escalation. Define escalation chain. Test regularly.
10	Memory usage slowly increasing over weeks	Memory leak detection	L2	Monitor heap usage over time. Compare before and after deployments. Profiler for root cause.
11	Need a single view of all service health	Service dashboard	L2	Grafana dashboard with 4 golden signals per service. Status page for stakeholders.
12	Incident happened but nobody documented what was done	Post-mortem process	L1	Blameless post-mortem within 48 hours. Timeline, root cause, action items. Track actions to completion.
13	Customer reports issue but we cannot reproduce it	Debug in production	L2	Check logs for their user_id/request_id. Trace the specific request through services.
14	Cost of monitoring tools is too high	Monitoring cost	L3	Sample data (not every event). Set retention tiers. Drop low-value metrics. Negotiate contracts.
15	New microservice deployed but no monitoring exists	Observability standards	L3	Standard: every service MUST expose metrics endpoint, structured logs, health check. Enforce via pipeline.

DOMAIN 8

Security & DevSecOps

Protecting systems and data

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	SonarQube found SQL injection in code	SAST finding	L1	Notify developer with specific file and line. Quality Gate blocks build. Developer must fix.
2	OWASP found Critical CVE in a dependency	SCA/CVE triage	L1	Create security ticket. Identify: which library, which version, what is the fix. Track SLA.
3	Developer wants to bypass security scan "just this once"	Policy enforcement	L1	Never bypass without security team approval. This is a compliance requirement.
4	Secrets found in GitHub repository	Secret exposure	L1	Rotate ALL exposed secrets immediately. Scan history. Add pre-commit hooks to prevent.
5	Container image has 47 known vulnerabilities	Image scanning	L2	Trivy/Grype scan in pipeline. Block images with Critical/High CVEs. Fix by updating base image.
6	Need to give app access to database without password in code	Secrets management	L2	AWS Secrets Manager, HashiCorp Vault. App reads secret at runtime. Never in code or config files.
7	Compliance audit asks: who deployed what and when	Audit trail	L2	Jenkins build history + Audit Trail plugin. Git commit history. All changes traceable.
8	Third-party library has no security patches available	Vulnerability exception	L2	Document: what CVE, why no fix, what mitigation. Security team approves. Re-review quarterly.
9	Need to prevent developers from disabling security stages	Pipeline security	L3	Shared Libraries with admin-only write. Security stages baked into template pipeline.
10	EC2 instance was compromised — how to respond	Incident response	L2	Isolate (remove from security group). Snapshot for forensics. Terminate and rebuild from AMI.
11	Need to implement least-privilege access across team	IAM design	L3	Role-based access. Per-team folders in Jenkins. Separate AWS accounts per environment.
12	SSL certificate is expiring in 3 days	Certificate management	L1	Renew via ACM (auto-renew) or manual renewal. Monitor expiry dates with alerts.
13	Need to scan running application for vulnerabilities	DAST	L3	OWASP ZAP or Burp Suite against running app. Finds runtime issues SAST misses. Run in staging.
14	Kubernetes secrets are stored as base64 (not encrypted)	K8s security	L2	Enable encryption at rest for etcd. Or use External Secrets Operator with Vault.
15	Need to restrict which container images can run in cluster	Admission control	L3	OPA/Gatekeeper or Kyverno. Policy: only images from approved registry. Block latest tag.

DOMAIN 9

Terraform & Infrastructure as Code

Defining infrastructure in code

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	Two engineers ran terraform apply simultaneously, state corrupted	State locking	L2	Use S3 backend with DynamoDB locking. Only one apply at a time. Remote state mandatory for teams.
2	Terraform wants to destroy a resource that should not be deleted	Lifecycle rules	L2	lifecycle { prevent_destroy = true }. Or use terraform state rm to remove from state without destroying.
3	Same infrastructure needed in dev, staging, production	Modules + workspaces	L2	Terraform modules for reusable components. Separate tfvars per environment.
4	terraform plan shows 50 changes but nothing was modified	State drift	L2	Someone changed infrastructure manually (ClickOps). terraform plan detects drift. Fix: import or revert.
5	Need to add an existing EC2 to Terraform management	Import	L2	terraform import aws_instance.my_ec2 i-1234567890. Writes real resource into state file.
6	Terraform state file contains secrets in plain text	State security	L2	Enable encryption on S3 backend. Restrict access to state bucket. Never commit state to Git.
7	Provider version changed and broke existing code	Version pinning	L2	required_providers { aws = { version = "~> 5.0" } }. Pin versions. Test upgrades.
8	terraform apply failed midway, resources partially created	Partial failure	L2	terraform apply again (idempotent). Or terraform destroy + apply. Check state for orphaned resources.
9	Need to share outputs between Terraform configurations	Remote state data	L3	data "terraform_remote_state" reads outputs from another state. Or use SSM Parameter Store.
10	Infrastructure code has no testing or validation	Terraform testing	L3	terraform validate, tflint, checkov for policy. Terratest for integration tests. Run in pipeline.
11	Terraform plan takes 10 minutes because state is huge	State performance	L3	Split into smaller state files by component (network, compute, database). Use -target for specific resources.
12	Need to manage infrastructure across multiple AWS accounts	Multi-account	L3	Provider aliases with assume_role. Or Terraform workspaces per account. AWS Organizations.

DOMAIN 10

Networking

How services talk to each other

#	Scenario (What You See)	Topic It Tests	Lvl	Key Concept
1	Application returns "Connection refused" on port 8080	Port connectivity	L1	Service not running, firewall blocking, listening on wrong interface (127.0.0.1 vs 0.0.0.0).
2	HTTPS certificate is invalid, browser shows warning	TLS/SSL	L1	Certificate expired, wrong domain, self-signed, incomplete chain. ACM or Let's Encrypt for valid certs.
3	Service A cannot reach Service B on internal network	VPC networking	L2	Check security groups (inbound on B allows A). Check route tables. Check NACLs.
4	DNS changes not propagating, old IP still used	DNS caching	L2	TTL on DNS records. Lower TTL before planned changes. Local cache: flush with systemd-resolve.
5	Load balancer health checks failing but app works	Health check config	L2	Check: health check path, port, timeout, interval. App must respond with 200 to health endpoint.
6	Intermittent "Connection timed out" errors	Network reliability	L2	Packet loss, DNS intermittent, TCP connection limits. Test with ping, traceroute, tcpdump.
7	Private EC2 needs to reach internet for package downloads	NAT Gateway	L2	EC2 in private subnet needs NAT Gateway in public subnet with route table entry.
8	Two services in different VPCs need to communicate	VPC Peering/TGW	L3	VPC Peering (simple, 1:1) or Transit Gateway (hub, many VPCs). Update route tables.
9	Application behind ALB, need to get real client IP	Proxy headers	L2	ALB adds X-Forwarded-For header. Application must read this header instead of connection source IP.
10	Port 443 open in security group but still cannot connect	Layered security	L2	Security group OK but NACL blocking? Route table missing? Instance not in correct subnet?
11	Microservice communication is slow between pods	Service mesh	L3	Istio/Linkerd for mTLS, traffic management, observability between services. Adds latency overhead.
12	Need to route api.company.com to one service, www to another	DNS/Ingress routing	L2	Route 53 CNAME to ALB. ALB host-based routing rules. Or K8s Ingress host/path rules.
13	WebSocket connections dropping after 60 seconds	Timeout settings	L2	ALB idle timeout default 60s. Increase for WebSocket. Also check proxy/nginx timeout.
14	Need to block traffic from specific countries	Geo-blocking	L3	AWS WAF with geo-match rules. CloudFront geo-restriction. Or application-level IP blocking.

Self-Assessment Checklist

Go through each domain. Count how many scenarios you could confidently handle. Be honest.

Linux (15 scenarios)	Foundation. Should know 12+ for any DevOps role. ____ / known
Git (14 scenarios)	Daily tool. Should know 10+ before applying. ____ / known
Jenkins (20 scenarios)	CI/CD core. Should know 12+ for L1 Jenkins roles. ____ / known
Docker (20 scenarios)	Container fundamentals. Should know 12+ for modern DevOps. ____ / known
Kubernetes (20 scenarios)	Container orchestration. Know 8+ for K8s roles, concepts for others. ____ / known
AWS (20 scenarios)	Cloud infrastructure. Should know 12+ for AWS-based roles. ____ / known
Monitoring (15 scenarios)	Observability. Should know 8+ for any operations role. ____ / known
Security (15 scenarios)	DevSecOps focus. Should know 10+ for DevSecOps roles. ____ / known
Terraform (12 scenarios)	IaC. Should know 6+ if Terraform is in the job description. ____ / known
Networking (14 scenarios)	Connectivity. Should know 8+ for any infrastructure role. ____ / known

The Architect Mindset

A beginner asks: "How do I use this tool?"

An engineer asks: "What breaks when this tool is under pressure?"

An architect asks: "How do I design so it never breaks?"

For every tool you learn, ask the 8 questions:

What happens when disk fills? When memory fills? When 100 users hit it? When it crashes at 2am? When credentials expire? When you upgrade? When you need to rollback? When the network fails?

Those 8 questions, applied to every tool, IS the architect mindset.

The scenarios that surprised you are your roadmap. Start there.